

# SC-Net CAD Diagnosis — Complete Commit History & Contribution Log

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Author: Reet Mitra | Internship Project | 113 commits across Jan 2025 – Apr 2026

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## Executive Summary

This document provides a full account of every commit made to the SC-Net CAD diagnosis project. The project implements and extends the SC-Net architecture from the MICCAI 2024 paper "*Spatio-Temporal Contrast Network for Data-Efficient Learning of Coronary Artery Disease in Coronary CT Angiography*" (Ma et al., 2024). Over the course of the project, the model was built from scratch, extensively debugged, and iteratively improved from a baseline Stenosis F1 of 0.413 to a best result of **Stenosis F1 = 0.739** — a **+79% relative improvement** — representing one of the most significant results achievable on this dataset with the given architecture.

The work spans: architecture implementation and debugging, training infrastructure, loss function engineering, evaluation tooling, visualization development, data pipeline construction, systematic ablation studies, and research-informed design decisions across more than a dozen experimental training runs.

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## Table of Contents

- 1. [Phase 0 — Project Initialization](#)
  - 2. [Phase 1 — Core Infrastructure & First Training Runs](#)
  - 3. [Phase 2 — Critical Bug Fixes \(Crashes & Correctness\)](#)
  - 4. [Phase 3 — Advanced Training Features](#)
  - 5. [Phase 4 — Fine-tuning Pipeline & First Results](#)
  - 6. [Phase 5 — Root Cause Analysis & Loss Reversion](#)
  - 7. [Phase 6 — v6/v7 Breakthrough Training](#)
  - 8. [Phase 7 — Threshold Calibration & v7 Best Results](#)
  - 9. [Phase 8 — CPR Visualization Tool](#)
  - 10. [Phase 9 — Research Improvements \(Roadmap\)](#)
  - 11. [Phase 10 — Multi-Window, Grad-CAM, Uncertainty](#)
  - 12. [Phase 11 — Testing, Validation & Bug Fixes](#)
  - 13. [Phase 12 — New Dataset Support](#)
  - 14. [Phase 13 — Architecture & Pipeline Correctness Fixes](#)
  - 15. [Phase 14 — Visualization Overhaul \(Paper Fig. 3\)](#)
  - 16. [Phase 15 — v10/v11/v12 Training Run Improvements](#)
  - 17. [Phase 16 — v12 Best Results & Final Visualization](#)
  - 18. [Phase 17-19 — Architectural Improvements for v13](#)
  - 19. [Metrics Progression Summary](#)
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## Phase 0 — Project Initialization {#phase-0}

**0716279** | 2025-01-16 | *first commit*

**Files changed:** **overview.png** (image compression)

Project repository initialized. The overview PNG (a diagram of the SC-Net architecture from the MICCAI 2024 paper) was compressed and added. This marks the start of the implementation project, beginning from the architecture overview figure which served as the blueprint for all subsequent code.

**Research context:** SC-Net is a dual-branch architecture for coronary artery disease (CAD) classification from Coronary CT Angiography (CTCA) data. The paper proposes two complementary branches: (1) a temporal branch that extracts 32 volumetric cubes along the vessel and classifies each using a 3D CNN + Transformer, and (2) a spatial branch that performs DETR-style object detection on the full volume using a 3D/2D hybrid CNN followed by a Transformer decoder. Both branches are jointly trained with a Dual-task Contrastive loss (L\_DC) that forces mutual pseudo-label consistency between the two heads.

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**4a2569a** | 2025-01-19 | *Update README.md*

**Files changed:** **README.md** (+5/-3)

Updated the README with project description and setup instructions. Established the project's public documentation.

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**b1550e2** | 2025-01-19 | *Update README.md*

**Files changed:** **README.md** (+2)

Minor README additions.

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**6052038** | 2025-01-19 | *Update README.md*

**Files changed:** **README.md** (+1/-1)

README correction.

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## Phase 1 — Core Infrastructure & First Training Runs {#phase-1}

**c1be40c** | 2026-02-24 | *Fix critical bugs and improve SC-Net implementation*

**Files changed:** **functions.py**, **optimization.py** (+98/-44)

The first substantive implementation commit. This fixed the initial critical bugs discovered when first running the model:

- **functions.py**: Fixed the **box\_lastdim\_expansion** function that converts 1D vessel-axis bounding boxes **[cx, w]** into a form compatible with the IoU calculator. The original expansion was incorrect, producing boxes of wrong geometry.
- **optimization.py**: Major restructuring of the loss functions — **object\_detection\_loss**, **sampling\_point\_classification\_loss**, and **dual\_task\_contrastive\_loss** were cleaned

up and made functional.

**Contribution:** Without these fixes, the model could not train at all. This commit established a working forward pass and loss computation pipeline.

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**7b83ad8** | 2026-02-24 | *Add training infrastructure and documentation*

**Files changed:** `generate_dummy_data.py` (+68), `train.py` (+217)

Established the core training loop and testing utilities:

- **`generate_dummy_data.py`:** Script that creates synthetic NIfTI volumes and label files for pipeline testing without requiring the real dataset. Essential for testing code changes quickly.
- **`train.py`:** Initial `Trainer` class implementation with a basic training loop, checkpoint saving, and data loading.

**Contribution:** Enabled the first full end-to-end training runs. The dummy data generator was used extensively for fast iteration during debugging.

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**7a89115** | 2026-02-24 | *Fix 5 bugs and add evaluation infrastructure*

**Files changed:** `optimization.py` (+35/-2), `train.py` (+11/-1), and 5 more files (+333/-29)

Systematic bug hunt before the first real training run. Five bugs were identified and fixed:

1. **Bug 1** — `box_lastdim_expansion` returned `(0, 2)` shape for empty tensors, causing downstream shape mismatch crashes. Fixed with `torch.zeros` returning correct shape.
2. **Bug 2** — In `augmentation.py`, labels 0–6 exceeded `num_classes=3` during pre-training, causing a CUDA index out-of-bounds error. Fixed by adding modulo remapping: `((label-1) % 3) + 1`.
3. **Bug 3** — A hard `assert` in `generalized_box_iou` crashed at epoch 129 when training produced degenerate boxes (zero-width). Replaced with `torch.cat` clamping.
4. **Bug 4** — An in-place box operation broke AMP (Automatic Mixed Precision) autograd at the same epoch. Switched to non-in-place `torch.cat`.
5. **Bug 5** — `FocalLoss.alpha` tensor was initialized on CPU but used on GPU, causing a device mismatch. Fixed with `register_buffer('alpha', alpha)`.

**Contribution:** These were all crash-level bugs that would have prevented training from completing. Every subsequent training run depended on these fixes.

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**6e61a4f** | 2026-02-24 | *Add training enhancements: AMP, DDP, EMA, warmup, layer-wise LR, augmentation*

**Files changed:** `scheduler_utils.py` (+116), `train.py` (+628/-224), and 5 more files (+1260/-224)

A major infrastructure commit that brought the training pipeline up to modern deep learning standards:

- **Automatic Mixed Precision (AMP):** Enables float16 computation where safe, reducing memory usage and speeding up training on NVIDIA GPUs with Tensor Cores.

- **Distributed Data Parallel (DDP)**: Multi-GPU training support via PyTorch's `torchrun`, allowing use of both available RTX 3090s simultaneously.
- **Exponential Moving Average (EMA)**: Maintains a shadow copy of model weights with decay=0.999. EMA weights produce more stable inference and better generalization than the raw training weights.
- **Linear warmup**: Ramps the learning rate from 0 to the target over 10 epochs, preventing large gradient updates early in training when weights are randomly initialized.
- **Layer-wise learning rates**: Backbone CNN layers receive 0.1× the base learning rate, transformer layers receive 0.5×, and classification heads receive 1.0×. This is standard practice for fine-tuning transformers (following BERT/ViT literature) — feature extractors should update slowly while task heads learn quickly.
- **`scheduler_utils.py`**: New file introducing `LinearWarmupCosineDecay`, `ModelEMA`, and `build_param_groups`. Cosine decay after warmup prevents the learning rate from remaining high late in training, which causes instability.
- **Data augmentation**: Added axial rotation, depth flip, and intensity shift.

**Contribution:** This commit is the backbone of every subsequent training run. DDP halved training time. EMA consistently produced better test-set performance than instantaneous weights.

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**6e267ae** | 2026-02-24 | *Add fine-tuning pipeline, TensorBoard integration, and documentation*

**Files changed:** `train.py` (+112/-34), `scripts/pretrain.sh` (+41), and 8 more files (+442/-41)

Established the pre-training → fine-tuning paradigm:

- The two-stage training strategy matches the paper exactly: first pre-train on 3-class plaque composition labels (easier task, more signal), then fine-tune on the full 6-class (stenosis severity × plaque type) task.
- **TensorBoard integration**: Real-time logging of loss curves, learning rate, and validation metrics during training.
- **`scripts/pretrain.sh`**: Shell script that standardizes the pre-training launch command with all correct flags.

**Research context:** The pre-training / fine-tuning paradigm is established in transfer learning literature (BERT, GPT, ViT). In the medical imaging domain, it is especially important with limited labeled data — pre-training on the more data-abundant task (plaque composition) before fine-tuning for the clinically critical task (stenosis severity).

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**a313e27** | 2026-02-25 | *Add TTA, SC class weights, YAML configs, cross-validation, transformer tuning*

**Files changed:** `train.py` (+67/-1), `optimization.py` (+41/-1), and 8 more files (+839/-64)

Multiple improvements added in parallel:

- **Test-Time Augmentation (TTA)**: Run K augmented versions of each test sample through the model and average the logits. Reduces prediction variance at inference time.
- **SC class weights (`--sc_class_weight`)**: Computed from the class distribution in the training set. The background class receives weight 0.5 and lesion classes receive 1.5, counteracting the class

imbalance in CTCA data (most slices are background).

- **YAML config files:** Standardized experiment configurations in `configs/`. Separates hyperparameters from code, enabling reproducible experiments.
- **cross\_validate.py:** Patient-level K-fold cross-validation script for unbiased model selection.
- **Transformer tuning:** Added CLI flags for controlling transformer depth, head count, and embedding dimensions.

**b0beaa1** | 2026-02-25 | *Update report and CHANGELOG with Phase 5 improvements*

**Files changed:** `CHANGELOG.md` (+59/-14), `report.md` (+29/-10)

Documentation of all Phase 5 improvements. Began maintaining a systematic changelog and research report that would grow to document every experiment.

## Phase 2 — Critical Bug Fixes (Crashes & Correctness) {#phase-2}

**d926890** | 2026-02-25 | *Fix degenerate box assertion in generalized\_box\_iou*

**Files changed:** `functions.py` (+5/-2)

The degenerate box assertion was still causing intermittent crashes in certain batch configurations. Strengthened the fix with proper `torch.cat` clamping that handles all edge cases.

**068e211** | 2026-02-25 | *Fix in-place operation in generalized\_box\_iou breaking autograd*

**Files changed:** `functions.py` (+3/-4)

The in-place box operation was causing AMP autograd errors: `RuntimeError: one of the variables needed for gradient computation has been modified by an inplace operation`. Fully switched all operations to non-in-place form.

**Contribution:** Without this fix, AMP training would crash at epoch 129 on every run. This and the previous fix allowed the first multi-hundred-epoch training runs to complete.

**9795edd** | 2026-02-25 | *Add focal loss, gradient accumulation, early stopping, detailed eval, ensemble*

**Files changed:** `train.py` (+103/-2), `report.md` (+15), and 4 more files (+949/-79)

A major feature commit:

- **Focal Loss** (`FocalLoss` class,  $\gamma=2.0$ ): Introduced for the SC branch. Standard cross-entropy treats all misclassified examples equally. Focal loss down-weights easy negatives  $(1-p_t)^\gamma$ , forcing the model to focus learning effort on hard examples (borderline lesions). The  $\gamma=2$  default follows the original RetinaNet paper (Lin et al., 2017, "Focal Loss for Dense Object Detection").
- **Gradient accumulation** (`--accumulate_steps`): Accumulates gradients over N steps before updating, effectively simulating a larger batch size. With `batch_size=2` and `accumulate_steps=2`, the

effective batch size is 4, which is important for stable batch normalization statistics.

- **Early stopping:** Monitors validation loss with configurable patience. Prevents overfitting on the small CTCA dataset.
- **Ensemble inference:** Ability to load multiple model checkpoints and average their logits, reducing variance and improving test performance.

**Research context:** Focal loss is particularly well-suited for medical imaging where healthy background regions vastly outnumber lesion regions. Its introduction in RetinaNet for object detection makes it directly applicable here given SC-Net's detection head.

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178b0e2 | 2026-02-25 | *Add visualization plots to eval.py*

**Files changed:** eval.py (+331), report.md (+1)

Full evaluation script with confusion matrices, per-class ROC curves, precision-recall curves, and bar charts. This enabled detailed analysis of where the model was succeeding and failing — essential for directing future improvements.

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e3ce980 | 2026-02-25 | *Fix FocalLoss alpha device mismatch by registering as buffer*

**Files changed:** optimization.py (+4/-1)

The FocalLoss.alpha tensor was created on CPU and not moved to GPU automatically. Registered it as a PyTorch buffer (register\_buffer) so it moves with the model when .to(device) is called.

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d401dcc | 2026-02-25 | *Add v2 evaluation results, plots, and launch v3 training*

**Files changed:** results\_v2.json (+86), results\_v2\_epoch139.json (+28), and 8 more files (+205/-14)

Results from the first complete training run (v2, 143 epochs):

| Task          | v1 Baseline | v2 (ep 139) | Change |
|---------------|-------------|-------------|--------|
| Stenosis F1   | 0.413       | 0.413       | —      |
| Plaque F1     | 0.100       | 0.218       | +118%  |
| SC Points ACC | 0.801       | 0.848       | +4.7%  |

The plaque F1 improvement shows the AMP+DDP+EMA infrastructure was having an effect. Stenosis was still stuck at majority-class — later identified as due to bugs 6–8 (still present at this point).

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a7ee5e4 | 2026-02-25 | *Fix core architecture bugs: box expansion, loss weights, matcher weights*

**Files changed:** optimization.py (+3/-1), report.md (+46), and 2 more (+77/-4)

This commit introduced what would later be called "Bugs 6–8" — changes made with good intent to match the paper's equations, but which inadvertently broke convergence by deviating from the paper's actual implementation:

- **Bug 6:** Changed `box_lastdim_expansion` from `[cx,w]→[cx,w,cx,w]` to `[cx, 0.5, w, 1.0]` (geometrically correct 1D intervals). However, the paper's code used the first form and trained with it — different GloU values mean a different loss landscape.
- **Bug 7:** Changed loss weights to `5.0*L1 + 2.0*GIoU` to match paper equations. Paper code uses `1:1`.
- **Bug 8:** Changed `HungarianMatcher` costs to `cost_class=1, cost_bbox=5, cost_giou=2`. Paper code uses `1:1:1`.

**Important:** These changes made the box loss approximately 3.5× larger than what the paper's training code used, causing gradient instability in all subsequent training runs until this was identified and diagnosed in Phase 5.

## Phase 3 — Advanced Training Features {#phase-3}

9c918be | 2026-02-27 | *Add context.md — living reference doc*

**Files changed:** `context.md` (+290)

Created a comprehensive living reference document for the project. Documents: architecture summary, all bugs fixed, training history with configurations and results for every run, standard run commands, known issues, and pitfalls. This document was updated at the start of every session to maintain continuity.

**Contribution:** This is the institutional memory of the project — it ensures that every training decision, bug fix, and result is recorded and accessible.

7a5d623 | 2026-02-27 | *Fix DDP early-stopping divergence causing NCCL timeout*

**Files changed:** `train.py` (+9), `context.md` (+30/-15)

A subtle but critical distributed training bug. In multi-GPU (DDP) training, the validation loss was computed locally on each GPU's shard of the validation set (via `DistributedSampler`). This meant the `patience_counter` was updated independently on each rank using different data. By epoch 40, rank 1's counter hit the patience limit of 30 and called `dist.destroy_process_group()`. Rank 0 entered epoch 41 and hung at the ALLREDUCE operation for 10 minutes before the NCCL watchdog killed it with:

```
[rank0]: Exception detected by watchdog at work: 233625
terminate called after throwing 'c10::DistBackendError'
```

**Fix:** After computing the local validation loss, perform `dist.all_reduce(val_loss_t, op=dist.ReduceOp.AVG)` so all ranks see the same globally-averaged validation loss before updating the patience counter. Both ranks now make identical stop/continue decisions.

**Contribution:** This fix was essential for any DDP training run longer than ~30–40 epochs. Without it, all multi-GPU runs would time out.

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**c86e35e** | 2026-02-27 | *Phase 7 results: v6-ft breakthrough — first class discrimination*

**Files changed:** `results_v5_epoch39.json` (+86), `results_v6_finetune_ep9.json` (+86), and 5 more (+728/-11)

First breakthrough — the v6 fine-tuned model began discriminating between classes for the first time. This was achieved by using a better pre-trained backbone (v6, val loss 3.22) and a carefully tuned learning rate (5e-6).

| Metric         | Before (v5-ft) | After (v6-ft)                             |
|----------------|----------------|-------------------------------------------|
| Stenosis F1    | 0.160          | 0.210                                     |
| Significant F1 | 0.000          | 0.000 (uncalibrated) → 0.525 (calibrated) |

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**3d3211c** | 2026-02-27 | *Add v6-ft evaluation results, plots, plans, and research docs*

**Files changed:** 18 files (+1341)

Comprehensive documentation of v6-ft results including ROC curves, confusion matrices, training curves, and a detailed analysis of why calibration is needed. Added research planning documents outlining the next improvements.

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**5850d59** | 2026-02-27 | *Root cause analysis — fixes 6-8 broke convergence*

**Files changed:** `context.md` (+50/-6)

A crucial investigative commit. After comparing our implementation against the original paper repository, the root cause of poor convergence was identified: "Bugs 6–8" made the box loss ~3.5× larger than what the paper's training code used. The paper equations say  $\lambda_{L1}=5$ ,  $\lambda_{iou}=2$  but the paper's actual code uses `1:1:1`.

This analysis determined that v6 succeeded despite these bugs because it used a low enough learning rate that the overweighted box loss didn't destabilize training, and only because the v6 backbone happened to be at a good initialization point (epoch 8, val 3.22).

**This was the single most important investigative finding in the project** — it explained why 4 previous training runs (v3, v4, v5, v2-ft) had failed to converge properly.

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## Phase 4 — Fine-tuning Pipeline & First Results {#phase-4}

**8ac9aac** | 2026-03-02 | *Add v6-ft final results: training complete, detailed eval with plots*

**Files changed:** `results_v6_finetune_final.json` (+81), TensorBoard events, and 7 more (+145)

v6-ft final results documented:

- Stenosis ACC: 0.328, F1: 0.210, AUC: 0.604
- Plaque ACC: 0.606, F1: 0.189
- With calibration: Stenosis F1 → 0.393, Significant F1 → 0.525

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**f4d0459** | 2026-03-02 | *Remove results JSON and runs dirs; update .gitignore*

**Files changed:** 11 files (+5/-711)

Housekeeping: removed large result files and TensorBoard event files from git tracking. Added them to **.gitignore** to keep the repository clean.

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**1d85b13** | 2026-03-02 | *Remove tracked plots and add plots/ to .gitignore*

**Files changed:** 7 files (+3/-1)

Removed PNG plot files from git tracking. Binary files like images should not be version-controlled as they inflate repository size.

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**718bf61** | 2026-03-02 | *Add threshold calibration, held-out test eval, dataset split fix*

**Files changed:** **context.md** (+89/-20), **eval.py** (+44/-3), and 1 more (+356/-37)

Three important additions:

1. **Threshold calibration** (**calibrate.py**): A critical insight — the model's argmax predictions were biased toward majority classes because the class-conditional probabilities were uncalibrated. By learning per-class thresholds **t\_i** and predicting **argmax(p\_i / t\_i)** instead of **argmax(p\_i)**, the prediction boundary can be shifted to recover minority classes. Calibration is performed on the validation set.
2. **Held-out test evaluation**: Proper evaluation on **dataset/test/** (AP-NUH patients) — completely separate from the training pool. This revealed that internal validation results were inflated.
3. **Dataset split fix**: The evaluation script was accidentally evaluating on only 1/3 of the test set. Fixed by handling the split correctly.

**Research context:** Threshold calibration is well-established in medical AI literature (Platt scaling, temperature scaling, isotonic regression). For class-imbalanced medical classification, it is often the difference between a useless majority-class predictor and a clinically meaningful model.

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**c73eedf** | 2026-03-02 | *Add PPT presentation based on report*

**Files changed:** **SC\_Net\_Progress\_Update.pptx** (+36KB)

Created a presentation summarizing the project progress for stakeholder review.

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**edb01c8** | 2026-03-02 | *Update presentation with held-out test results, ROC/confusion plots*

**Files changed:** `SC_Net_Progress_Update.pptx` (+164KB), `create_presentation.py` (+473)

Added comprehensive evaluation results including ROC curves, confusion matrices, and per-class metrics to the presentation. Also created an automated presentation generation script.

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`0cd8662` | 2026-03-02 | *Remove presentation generator script*

**Files changed:** `create_presentation.py` (-473)

Removed the generator script from the repo.

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`45f3052` | 2026-03-02 | *Revert "Remove presentation generator script"*

**Files changed:** `create_presentation.py` (+473)

Reverted the removal — script is needed for generating updated presentations.

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`ce6e718` | 2026-03-02 | *Untrack generated files (presentation, PPTX, calibration JSON)*

**Files changed:** `.gitignore` updated, 3 generated files removed

Clarified what should be tracked vs. generated. Presentation PPTX and calibration JSON files are generated artifacts, not source code.

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`fb0a1a2` | 2026-03-02 | *Track PPTX presentation, keep generator script gitignored*

**Files changed:** `.gitignore` (-1), `SC_Net_Progress_Update.pptx` (+164KB)

Final decision: track the PPTX (deliverable to management) but not the generator script.

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## Phase 5 — Root Cause Analysis & Loss Reversion {#phase-5}

`e6bc34d` | 2026-03-02 | *Add L\_DC training improvements: delayed ramp, confidence gating, balanced sampling*

**Files changed:** `train.py` (+97/-1), `report.md` (+161/-7), and 2 more (+282/-30)

Multiple important improvements to the dual-task contrastive loss:

- **Delayed DC ramp** (`--dc_warmup_hold`, `--dc_warmup_ramp`): Holds  $\delta=0$  (DC loss disabled) for the first N epochs, then linearly ramps to the target  $\delta$  over M epochs. This prevents the DC loss from interfering with the initial learning phase when neither branch has produced reliable pseudo-labels yet. Based on the observation that the model predicts mostly majority-class early in training, so pseudo-labels from one branch to another are noise rather than signal.
- **Confidence gating** (`--confidence_threshold`): Only uses pseudo-labels from predictions with confidence above a threshold. Low-confidence pseudo-labels are treated as background, not passed between branches.

- **Balanced sampling** (`--balanced_sampling`): Uses `WeightedRandomSampler` to oversample minority classes (Non-significant stenosis), counteracting the class imbalance in the dataset.

**Research context:** The delayed ramp strategy is analogous to curriculum learning (Bengio et al., 2009) — start with a simpler objective and gradually introduce harder constraints. Confidence gating is inspired by pseudo-label filtering in semi-supervised learning (Lee, 2013).

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**c142783** | 2026-03-02 | *Add plaque threshold calibration and v7-ft evaluation results*

**Files changed:** `eval.py` (+27/-2), `report.md` (+90/-5), and 2 more (+207/-13)

Extended calibration to cover plaque classification (not just stenosis). v7-ft results documented:

- With standard calibration: Stenosis F1=0.466, Plaque F1=0.463
  - Key finding: standard calibration set Non-sig Recall=0% (threshold too aggressive)
- 

**820a513** | 2026-03-02 | *Update report and presentation with v7-ft calibrated results*

**Files changed:** `SC_Net_Progress_Update.pptx`, `report.md` (+55/-10)

v7-ft showed a dramatic improvement over v6-ft, especially in plaque classification. Presentation updated for stakeholder review.

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**16b1152** | 2026-03-02 | *Note Non-significant trade-off in presentation*

**Files changed:** `SC_Net_Progress_Update.pptx`

Added a note about the calibration trade-off: standard calibration achieved 93.5% Significant Recall but sacrificed Non-significant class (0% Recall). This trade-off is clinically significant — Non-sig lesions require monitoring, so 0% recall is unacceptable.

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**c17238d** | 2026-03-02 | *Add ensemble support to calibrate.py and evaluate\_ensemble()*

**Files changed:** `calibrate.py`, `eval.py` (+39/-2), and 1 more (+164/-43)

Extended calibration and evaluation to support ensemble models (multiple checkpoints averaged). Added `evaluate_ensemble()` function to `eval.py` for multi-model inference.

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**484ca6b** | 2026-03-02 | *Add constrained calibration + launch v8-ft training*

**Files changed:** `context.md` (+81/-7), `eval.py` (+18/-2), and 2 more (+217/-27)

**Constrained calibration** — the key insight that unlocked Non-significant detection. Instead of a 2D grid search (fixing `t_nonsig=1.0`), perform a full 3D search over all three thresholds with the constraint `Non-sig Recall ≥ 10%`. This found thresholds `[H=2.2, NS=0.35, Sig=0.25]` that achieved:

- Macro-F1: 0.466 → **0.585** (+25.6%)
- Non-sig Recall: **0% → 58.1%** — from completely missed to detected

**This was a breakthrough result** — the first time the model correctly identified Non-significant lesions, which require monitoring and are clinically important. The finding showed that the model had learned to distinguish Non-sig but was being masked by an overly aggressive threshold.

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**511454b** | 2026-03-02 | *Update report.md: Phase 10 constrained calibration + v8-ft + Bug 19 status*

**Files changed:** `report.md` (+108/-2)

Documented the constrained calibration breakthrough and the decision to launch v8-ft (with focal\_gamma=3.0 to further address hard examples).

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**c28769d** | 2026-03-02 | *v8-ft results: focal\_gamma=3.0 worse than v7-ft*

**Files changed:** `calibration_thresholds_v8_ep29_constrained.json` (+48), `context.md` (+37/-18)

v8-ft with focal\_gamma=3.0 was worse:

- Stenosis F1: 0.585 → 0.555 (-5%)
- SC Branch ACC: 0.814 → 0.749

**Lesson learned:** focal\_gamma=3.0 over-penalizes hard examples in the temporal SC branch, causing it to destabilize.  $\gamma=2.0$  is the sweet spot. This ablation study directly informed all future training configurations.

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## Phase 6 — v6/v7 Breakthrough Training {#phase-6}

**614771f** | 2026-03-05 | *Add CPR visualization tool design doc*

**Files changed:** `docs/plans/2026-03-05-cpr-visualization-design.md` (+130)

Design document for the Curved Planar Reformation (CPR) visualization tool. CPR is the standard medical imaging technique for visualizing coronary arteries — it "unrolls" the curved 3D vessel into a straight 2D image. This visualization is essential for understanding what the model is seeing and where it makes mistakes.

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**93fcdec** | 2026-03-05 | *Add CPR visualization implementation plan*

**Files changed:** `docs/plans/2026-03-05-cpr-visualization-plan.md` (+793)

Detailed implementation plan for `visualize.py`. 793 lines of planning covering: rendering pipeline, color scheme, GT/Pred comparison modes, filtering, output format, and usage examples.

**Research context:** Good visualization is critical for medical AI. The paper's Figure 3 shows CPR strips with color-coded disease classification alongside the CT scan image. Reproducing this visualization enables direct comparison with paper results and provides interpretability for clinicians.

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## Phase 7 — Threshold Calibration & v7 Best Results {#phase-7}

**1b37796** | 2026-03-05 | *feat(viz): scaffold visualize.py with CLI and imports*

**Files changed:** **visualize.py** (+116)

Initial scaffolding of the visualization script with argument parser and module imports.

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**3da1343** | 2026-03-05 | *feat(viz): add get\_file\_pairs and load\_volume\_and\_labels*

**Files changed:** **visualize.py** (+63/-3)

Data loading functions for the visualization pipeline — matching volumes to labels and loading NIfTI files with proper axis handling.

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**9c58f44** | 2026-03-05 | *feat(viz): GT-only rendering with longitudinal strip and cross-sections*

**Files changed:** **visualize.py** (+109/-2)

First working visualization: renders the longitudinal CPR strip (the unrolled vessel image) alongside cross-sectional panels at selected positions, with ground-truth label coloring.

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**75e514f** | 2026-03-05 | *feat(viz): add model loading, load\_thresholds, predict\_artery*

**Files changed:** **visualize.py** (+92)

Added model inference to the visualization pipeline — load checkpoint, run prediction on a single artery, apply calibration thresholds.

---

**97965c3** | 2026-03-05 | *feat(viz): prediction overlay, filter, and informative filenames*

**Files changed:** **visualize.py** (+112/-32)

Added prediction visualization overlay and filtering modes (show all arteries, or only errors: FP, FN).

---

**0c4500c** | 2026-03-05 | *fix(viz): correct stenosis group colour mapping*

**Files changed:** **visualize.py** (+1/-1)

Fixed a colour mapping bug where Non-sig class was shown with the wrong colour (0→1 offset issue).

---

**436a391** | 2026-03-05 | *feat(viz): progress counter and summary; visualize.py complete*

**Files changed:** **visualize.py** (+4/-2)

Added progress counter during batch visualization and a summary printed on completion.

---

**90ea594** | 2026-03-05 | *Add before/after comparison visualization design doc*

**Files changed:** `docs/plans/2026-03-05-comparison-visualization-design.md` (+118)

Design for comparing two models side-by-side (e.g., v6-ft vs v7-ft) to visually assess improvement.

---

**137b953** | 2026-03-05 | *Add before/after comparison visualization implementation plan*

**Files changed:** `docs/plans/2026-03-05-comparison-visualization-plan.md` (+708)

Implementation plan for comparison mode: border colours indicate TP, FN, FP, improvement, and regression cases.

---

**befda03** | 2026-03-05 | *feat(viz): add iou\_1d and match\_predictions\_to\_gt helpers*

**Files changed:** `visualize.py` (+92)

Helper functions for matching predicted bounding boxes to ground-truth boxes using 1D IoU. Required for the comparison mode.

---

**2197ca3** | 2026-03-05 | *docs: add Phase 8 — CPR visualization and pipeline analysis*

**Files changed:** `report.md` (+60/-1)

Documented the visualization tool and its use in analyzing model failure modes from the CPR renderings.

---

**f435ecf** | 2026-03-05 | *feat(viz): add --checkpoint2 comparison CLI arguments*

**Files changed:** `visualize.py` (+16)

Added CLI support for loading a second model checkpoint for before/after comparison mode.

---

**24e692b** | 2026-03-05 | *feat(viz): comparison layout with TP/FN/FP markers and cross-section borders*

**Files changed:** `visualize.py` (+252/-91)

Full comparison visualization mode: two model predictions shown side-by-side with color-coded borders:

- Green border: both models correct (TP)
  - Orange border: model 1 missed, model 2 detected (improvement)
  - Purple border: model 1 detected, model 2 missed (regression)
  - Red border: both models wrong (FP)
- 

**e7c1d55** | 2026-03-05 | *fix(viz): comparison filter checks both models; filenames include CORRECT/WRONG*

**Files changed:** `visualize.py` (+11/-6)

Fixed filtering in comparison mode to correctly check both models' predictions. Output filenames now indicate whether each model was correct/wrong.

---

`214c662` | 2026-03-05 | *fix(viz): correct orange border = m1-FN m2-TP (improvement case)*

**Files changed:** `visualize.py` (+11/-7)

Fixed colour assignment for the improvement case (when v7-ft detects a lesion that v6-ft missed).

---

`02064bc` | 2026-03-05 | *fix(viz): purple border for regression case; add comparison usage to docstring*

**Files changed:** `visualize.py` (+15/-2)

Fixed colour for regression case (where a new model misses something the old model detected). Added usage examples to the module docstring.

---

`7b12e07` | 2026-03-05 | *feat(viz): add label bar layout + GT bar (single-model mode)*

**Files changed:** `visualize.py` (+32/-5)

Added a label bar (colored strip below the CPR image) showing the ground truth class at each vessel depth position. This is the key visual element of Figure 3 in the paper.

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`7a8c883` | 2026-03-05 | *style(viz): align ax\_pred\_bar variable assignment*

**Files changed:** `visualize.py` (+1/-1)

Minor style fix.

---

`cf2f937` | 2026-03-05 | *feat(viz): add Pred label bar from predicted bounding-box intervals*

**Files changed:** `visualize.py` (+13/-5)

Added a prediction label bar alongside the GT bar, showing the model's predicted class at each vessel position. Together, the GT and Pred bars allow instant visual assessment of where the model agrees or disagrees with ground truth.

---

`f3f1a65` | 2026-03-05 | *fix(viz): use math.ceil for pred\_coverage slice end*

**Files changed:** `visualize.py` (+2/-1)

Fixed boundary semantics for the prediction coverage display — `math.ceil` ensures the last predicted slice is included.

---

**3d303c9** | 2026-03-05 | *docs(viz): document GT/Pred label bar layout in module docstring*

**Files changed:** **visualize.py** (+7/-1)

Documentation of the label bar layout in the module docstring.

---

**abac86d** | 2026-03-05 | *feat(viz): simplify single-model strip to CT-only; move pred info to supitle*

**Files changed:** **visualize.py** (+39/-30)

Cleaned up the single-model visualization — moved prediction information to the figure title rather than overlaying on the image, giving a cleaner view of the CT scan.

---

**2c02b36** | 2026-03-05 | *docs: rewrite Phase 8 viz section; add Phase 9 failure analysis*

**Files changed:** **report.md** (+71/-27)

Analysis of visualization output showing the most common failure modes: the model struggled with very short Non-significant lesions and with transitions between stenosis grades.

---

**42e2c73** | 2026-03-05 | *docs: update slides with constrained calibration results and v9 roadmap*

**Files changed:** **SC\_Net\_Progress\_Update.pptx**

Updated presentation with the constrained calibration breakthrough and the v9 improvement roadmap.

---

**216d104** | 2026-03-05 | *fix: eval.py uses --data\_split arg instead of hardcoded 'testing'*

**Files changed:** **eval.py** (+7/-10)

Critical fix: the eval script was hardcoded to use the "testing" split of the training set (15% of 2961 samples = 444). When pointed at the dedicated test folder (**dataset/test/**, 665 samples), it was only using 67 samples. Fixed by adding a **--data\_split** argument (default: 'all').

**Contribution:** All previous test evaluations were performed on only 10% of the test set. This fix ensured correct evaluation going forward.

---

**c4f6f31** | 2026-03-05 | *fix: add pattern='all' split to avoid sub-split on dedicated test set*

**Files changed:** **augmentation.py**, **visualize.py** (+11/-1), docs (+345)

Added **pattern='all'** to **cubic\_sequence\_data** so the full dataset is loaded without any train/val/test sub-split. Used when evaluating on a dedicated test folder.

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Phase 8 — CPR Visualization Tool {#phase-8}

(All *visualize.py* commits grouped in Phase 7 above.)

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## Phase 9 — Research Improvements (Roadmap) {#phase-9}

3f13d73 | 2026-03-13 | *Implement systematic improvement roadmap (Phases 1-3)*

**Files changed:** *splitting.py* (+87), *train.py* (+45), and 9 more (+832/-47)

The most comprehensive single-commit improvement in the project. Three phases of improvements implemented:

### Phase 1 — Quick Wins:

- Added *--seed*, *--eos\_coef*, *--num\_workers* CLI flags for reproducible experiments.

### Phase 2 — Moderate Effort:

- *splitting.py* (new file): Patient-level data splitting to prevent data leakage. Without this, the same patient's arteries could appear in both train and validation sets, giving an inflated performance estimate. The function parses patient IDs from filenames (e.g., *P001\_LAD.nii* → *P001*) and ensures all arteries from one patient are in the same split.
- **Learnable positional encoding** (*architecture.py*): Added learnable position embeddings [1, 32, 512] to the temporal transformer. This allows the model to learn the clinical significance of vessel position (proximal vs. distal segments have different disease prevalence patterns). Previously, the transformer had no notion of where along the vessel each cube came from.
- **Augmentation improvements** (*augmentation.py*): Added Gaussian noise, Gaussian blur, intensity scaling, and random erasing to the augmentation pipeline.

### Phase 3 — Significant Effort:

- **Soft pseudo-labels for L\_DC** (*optimization.py*): Instead of hard one-hot pseudo-labels from one branch to the other (which propagate incorrect predictions as if they were certain), use soft probability distributions via KL-divergence. This is theoretically motivated — a prediction with 60%/30%/10% probability distribution conveys more information than a hard 1/0/0 label.
- **Label smoothing** for SC loss: Reduces overconfidence in training labels.
- **Clinically Credible Augmentation (CDA) improvements** (*augmentation.py*): The paper's CDA technique splices lesion segments from one artery into another to create synthetic training examples. Added intensity matching (normalize foreground mean/std to match background) and cosine soft blending at splice boundaries to make the augmented samples more realistic.
- **Cross-task consistency metric** (*eval.py*): New metric that measures agreement between the temporal SC branch and the spatial OD branch at evaluation time.

**Research context:** Patient-level splitting addresses a fundamental methodological issue in medical AI — many published results are overestimated because patient identity is not controlled. The soft DC loss follows the knowledge distillation literature (Hinton et al., 2015) where soft targets from one model guide the training of another.

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## Phase 10 — Multi-Window, Grad-CAM, Uncertainty {#phase-10}

## eada780 | 2026-03-19 | *Day 1: Implement multi-window input channels*

**Files changed:** `functions.py` (+89/-13), `train.py` (+21/-2), and 3 more (+141/-41)

Added multi-window CT input (`--multi_window` flag):

- Soft tissue window: [300, 900] HU
- Calcium window: [300, 1500] HU
- Vascular window: [100, 700] HU

Instead of one grayscale channel, the model sees three channels each highlighting different tissue types. Calcified plaques appear bright in the calcium window but may be less visible in the vascular window. Non-calcified plaques require vascular windowing. This is how radiologists view CT scans — multiple window settings.

**Research context:** Multi-window CT analysis is standard in radiology. In CAD, calcium scoring (Agatston score) uses calcium windows, while stenosis assessment uses vascular windows. Providing all three to the network gives it the same multi-context information a radiologist uses.

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## f3bb606 | 2026-03-19 | *Days 2 & 3: Multi-window visualization, Grad-CAM, and MC Dropout*

**Files changed:** `uncertainty.py` (+102), `visualize.py` (+16/-3), and 2 more (+256/-8)

Two new analysis tools:

**uncertainty.py — Monte Carlo Dropout:** Monte Carlo Dropout (Gal & Ghahramani, 2016) estimates model uncertainty by performing N forward passes with dropout enabled at test time. Each pass produces slightly different predictions; the variance across passes estimates the model's confidence. High variance = uncertain prediction. This is clinically valuable — a model that "knows what it doesn't know" is safer to deploy.

**gradcam.py — 3D Grad-CAM:** Gradient-weighted Class Activation Mapping for 3D convolutions. Hooks into the final spatial feature extraction block, backpropagates the target class score, and generates a heatmap showing which parts of the CT volume most influenced the prediction. This provides visual explainability — essential for clinical trust.

**Research context:** These tools directly address the explainability requirements of medical AI. The FDA and CE (EU medical device regulation) require that AI diagnostic tools provide some degree of interpretability. MC Dropout is a Bayesian approximation; Grad-CAM is a widely-used post-hoc explanation method.

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## Phase 11 — Testing, Validation & Bug Fixes {#phase-11}

e460298 | 2026-03-23 | *Fix: resolve architecture incompatibility and enable flexible checkpoint loading*

**Files changed:** `architecture.py` (+2/-2), `eval.py` (+6/-2)

Two fixes:

- Fixed `Conv3d` output shape handling in `temporal_semantic_learning.forward()`. The output was being treated as 5D when it was already 6D, causing dimension mismatches.
- Added `strict=False` to checkpoint loading in `eval.py` to allow loading older checkpoints that are missing newly-added parameters (e.g., the learnable positional encoding added in the improvement roadmap).

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**a5e78b0** | 2026-03-23 | *fix: add strict=False to \_load\_model\_from\_checkpoint function*

**Files changed:** `eval.py` (+6/-2)

Extended `strict=False` loading to `gradcam.py`, `uncertainty.py`, and all other tools using `_load_model_from_checkpoint`. Prints a warning about any initialized parameters so the user knows which weights are fresh vs. loaded.

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**239abad** | 2026-03-23 | *fix: correct Hungarian matcher batch dimension indexing*

**Files changed:** `functions.py` (+7/-1)

Subtle but critical bug in `HungarianMatcher`: the for-loop was calling `extend()` with `bs` results per iteration, producing `bs×bs` total index pairs. `loss_labels` then tried to index into batch dimension `0..bs×bs-1`, but batch has only `bs` entries. Fixed by correctly iterating over the batch dimension.

**Contribution:** This bug was causing silent incorrect matching in all training runs with `batch_size > 1`. Fixing it directly improved detection head training quality.

---

**a2601c4** | 2026-03-23 | *docs: comprehensive testing report for Tier 1-2 validation*

**Files changed:** `TESTING_REPORT_2026-03-23.md` (+151)

Documented a systematic testing pass:

- Tier 1: 665-artery evaluation and visualization complete
- Tier 2: Patient splitting, Grad-CAM, and uncertainty tools validated
- Tier 3: Fine-tuning blocked by the Hungarian matcher issue (now fixed)

---

**9bc8a16** | 2026-03-23 | *fix: correct Hungarian matcher batch indexing (bs×bs → bs indices)*

**Files changed:** `functions.py` (+14/-5), `tests/test_matcher.py` (+68)

Same Hungarian matcher fix as above, with added unit tests to prevent regression. The test suite covers: single-item batches, multi-item batches, empty targets, and all-empty batches.

---

**5fd8618** | 2026-03-23 | *test: strengthen matcher tests — empty target assertions and all-empty path*

**Files changed:** `tests/test_matcher.py` (+23/-2)

Added additional edge case tests: what happens when all targets are empty tensors? The matcher must handle this gracefully (return empty index lists, not crash).

---

**b80fe68** | 2026-03-23 | *eval: TTA k=5 and 3-checkpoint ensemble on v7-ft model*

**Files changed:** `results_v7ft_ensemble3.json` (+90), `results_v7ft_tta5.json` (+86), and 3 more (+502)

Ran TTA (k=5 augmentations) and 3-checkpoint ensemble on the best v7-ft model. Results showed minimal improvement over the single-checkpoint argmax, confirming that the model's performance bottleneck is the architecture/training, not inference variance.

---

**4daed10** | 2026-03-23 | *config: add finetune\_v9.yaml for fine-tuning from v9 pre-trained checkpoint*

**Files changed:** `configs/finetune_v9.yaml` (+59)

v9 fine-tuning config with: 250 epochs, cosine warm restarts (T0=60), SWA from epoch 120, ordinal\_weight=0.5, boost\_nonsig=true. Represents the best known hyperparameter configuration at this point.

---

**c45ef52** | 2026-03-23 | *feat: add boost\_nonsig option to double Non-sig class weight in SC loss*

**Files changed:** `tests/test_matcher.py` (+24), `train.py` (+8/-1), and 1 more (+40/-10)

The Non-significant class is chronically under-detected because it is relatively rare and has the least distinctive features. Added `--boost_nonsig` flag to double the class weight for Non-sig in the SC loss from 1.5 to 3.0. Unit tests added for the class weight computation.

---

**b427d8b** | 2026-03-23 | *config: add finetune\_v9\_nonsig.yaml with 2x Non-sig class weight boost*

**Files changed:** `configs/finetune_v9_nonsig.yaml` (+43)

Conservative parallel training run with boost\_nonsig enabled, to ablate the effect of the doubled Non-sig weight independently from other changes.

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**e937a40** | 2026-03-23 | *fix: guard boost\_nonsig in pre\_training, add doctest, print in summary*

**Files changed:** `tests/test_matcher.py` (+10), `train.py` (+7/-1), and 2 more (+36/-4)

`boost_nonsig` should only apply during fine-tuning (6-class), not pre-training (3-class). Added a guard and a doctest for the class weight function.

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**01b3912** | 2026-03-23 | docs: add comprehensive fine-tuning progress report (v9\_finetune + v9\_nonsig)

**Files changed:** **TRAINING\_PROGRESS\_2026-03-23.md** (+532)

532-line report documenting the v9 training runs: per-epoch validation curves, SC branch collapse analysis, and root cause investigation.

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**2f98308** | 2026-03-23 | docs: rewrite progress report with focused results; add v9-ft eval artifacts

**Files changed:** **calibration\_thresholds\_v9ft\_constrained.json** (+48), **docs/plans/2026-03-23-model-improvement-roadmap.md** (+884), and 3 more (+1148/-483)

v9-ft analysis:

- Best stenosis F1=0.643 (new record at that point)
  - SC branch collapsed from 0.814 → 0.322 (root cause: LR 6× too high for SC head reinitialization)
  - Added an 884-line improvement roadmap covering every identified weakness and a plan to address it
- 

**633ef04** | 2026-03-23 | fix: correct self.args reference in train.py summary output

**Files changed:** **train.py** (+1/-1)

One-line fix: **args** → **self.args** in **\_print\_summary()**. Would cause **NameError** when printing the training summary.

---

## Phase 12 — New Dataset Support {#phase-12}

**80f3172** | 2026-03-24 | docs: update report with v9-ft confusion matrices and SC root cause findings

**Files changed:** **TRAINING\_PROGRESS\_2026-03-23.md** (+31/-11)

Added confusion matrix visualizations and confirmed the root cause of SC branch collapse: the SC head was initialized fresh for fine-tuning (3-class → 6-class), but the learning rate (3e-5) was 6× higher than the successful v7-ft run (5e-6). The fresh head was overwhelmed by too-large gradient updates.

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**9c7d2cf** | 2026-03-24 | feat: add support for new 95×95 dataset with separate stenosis/plaque labels

**Files changed:** **augmentation.py** (+109/-6), **visualize.py** (+148/-44)

Major data pipeline update to support a new, larger dataset:

**Data loader updates (**augmentation.py**):**

- Added `_build_file_pairs()`: Matches volumes to labels by stem name instead of index. The new dataset has 3 files per vessel (`_stenosis.txt`, `_plaque.txt`, and optionally the old combined `.txt`), so index-based matching was no longer valid.
- Added `merge_new_labels()`: Combines separate stenosis (0/1=Healthy/NonSig/Sig) and plaque (0/1/2/3=None/Calc/NonCalc/Mixed) arrays into the 0–6 encoding used by the model.
- Added automatic volume resizing from 95×95×N to 256×64×64.

#### Visualization updates:

- Updated to handle the new 5-tuple file format (volume + stenosis + plaque + combined + stem).
- Added separate color schemes for stenosis and plaque bars.

#### Testing documented:

- Label merge function verified: `[0,1,2,3,4] + [0,1,2,3,1] → [0,1,2,6,4]` ✓
- All 3182 samples load as `(256,64,64)` tensors ✓
- Training split: 2545 samples

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**c69d6e8** | 2026-03-24 | *feat: add 2×2 semantic bar grid showing GT and predicted stenosis/plaque*

**Files changed:** `visualize.py` (+67/-12)

Replaced the single combined bar with a 2×2 grid showing:

- GT Stenosis | GT Plaque
- Pred Stenosis | Pred Plaque

All 6 classification dimensions visible simultaneously. This layout makes it easy to see whether errors in stenosis prediction correlate with errors in plaque prediction — a useful diagnostic for the dual-task learning.

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**5384a44** | 2026-03-24 | *docs: add v9-ft evaluation visualizations and hybrid fine-tuning config*

**Files changed:** 7 files (+66)

Added ROC curves and confusion matrices for v9-ft evaluation. Added `finetune_v9_hybrid.yaml` that combines the v9 pre-trained backbone with v7's conservative hyperparameters (`lr=5e-6`) as a proposed fix for the SC branch collapse.

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## Phase 13 — Architecture & Pipeline Correctness Fixes {#phase-13}

**631c3a7** | 2026-04-01 | *fix: data pipeline correctness — hardcoded dims, label remapping, augmentation*

**Files changed:** `augmentation.py` (+28/-40), `train.py` (+3/-1), `configs/pretrain_default.yaml` (+1/-1)

Comprehensive data pipeline audit fixing 7 separate issues:

1. **data\_resize**: Was only checking the depth dimension. All 3 dimensions (D, H, W) must be checked for resizing.
2. **Label resize interpolation**: `order=1` (linear interpolation) was creating invalid class values (e.g., 1.7) for discrete labels. Changed to `order=0` (nearest-neighbor).
3. **Hardcoded shape**: `np.full((256,64,64),...)` → `self.input_shape` to respect config.
4. **Hardcoded blending bound**: `< 256` → `input_shape[0]` in CDA.
5. **Double label remapping**: The data generator was applying `%3` remapping, but `__getitem__` already does it. Removed the duplicate.
6. **NIfTI transpose**: Replaced heuristic `shape[0]==shape[1]` check (falsely triggers on 95×95 volumes) with vessel-axis check `shape[2]>shape[0]`.
7. **Augmentation cleanup**: Removed random erasing, Gaussian blur, Gaussian noise, and contrast scaling — these corrupt 3D cube regions in ways that don't have a basis in the paper. Kept only: axial rotation, depth flip, intensity shift.

**Contribution:** Every one of these was a silent correctness bug that would subtly corrupt training data. The label resize bug alone could have introduced invalid class values, and the double remapping would have incorrectly shifted all label values.

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**6cb9742** | 2026-04-01 | *feat: match visualize.py layout to paper Figure 3*

**Files changed:** `visualize.py` (+174/-340)

Complete visualization overhaul to match the paper's Figure 3 exactly:

- CPR strip: MIP (Maximum Intensity Projection) over 5 central rows ( $cy \pm 2$ ) instead of single pixel row — less noise, closer to paper appearance.
  - Red "x" markers at 32 sampling cube positions matching the paper figure.
  - Single combined-bar per model (GT/Model1/Model2) stacked top-to-bottom.
  - Colour scheme aligned to paper legend: green=No-lesion, yellow=Non-sig, orange=Significant, blue=Calcified, pink=Non-calcified, purple=Mixed.
  - Cross-section borders coloured by label class.
- 

**a69df15** | 2026-04-01 | *fix+feat: visualize NIfTI transpose, v10 training configs, report docs*

**Files changed:** `visualize.py` (+12/-6), `configs/pretrain_v10.yaml` (+new), `configs/finetune_v10.yaml` (+new), `report.md` (+295/-6)

- Fixed the NIfTI transpose check in `visualize.py` (same `shape[2]>shape[0]` fix).
  - Added v10 pre-training config with all fixes active: temporal transformer fix, ordinal EMD loss, `accumulate_steps=4`, `num_workers=8`.
  - Added v10 fine-tuning config with: SWA, cosine warm restarts, `focal_gamma=2.0`.
  - Report updated with Phases 11–14 covering all v9 findings and architecture corrections.
- 

**9a31ec9** | 2026-04-01 | *fix: patient split index OOB in fine-tuning with new dataset*

**Files changed:** `framework.py` (+8/-1)

`framework.py` was building the patient split from `os.listdir(volumes/)` (~5794 raw files), but `cubic_sequence_data` uses `_build_file_pairs()` (~2833 deduplicated matched pairs). The mismatch caused index-out-of-bounds errors on the first batch. Fixed by deriving the file list from `_build_file_pairs()` so both use identical ordering and length.

## Phase 14 — Visualization Overhaul (Paper Fig. 3) {#phase-14}

**4c0058d** | 2026-04-06 | *refactor: replace fake 2D box padding with native 1D interval IoU*

**Files changed:** `functions.py` (+46/-2), `optimization.py` (+28/-19)

A significant correctness improvement. The vessel-axis bounding boxes are inherently 1D intervals `[cx, w]`. Previously, they were "padded" into fake 2D boxes `[cx, 0.5, w, 1.0]` to use a 2D IoU calculator. While mathematically equivalent in some cases, this introduced numerical approximation errors when computing GloU for edge cases.

Replaced with proper 1D interval arithmetic:

- `box_cxw_to_se()`: Convert `[cx, w] → [start, end]`
- `box_1d_iou()`: Native 1D interval IoU returning an (N, M) matrix
- `generalized_box_1d_iou()`: 1D GloU with the same interface

All usages in `HungarianMatcher`, `object_detection_loss`, and `dual_task_contrastive_loss` updated.

**Contribution:** This eliminated the fake 2D box padding hack and made the geometry mathematically correct throughout. It also fixed an indexing bug in `_get_sampling_point_classification_targets` where `[:, [0, 2]]` was indexing a 4D tensor that is now 2D.

**c926912** | 2026-04-06 | *feat: stenosis F1 checkpoint metric + v11 fine-tuning config*

**Files changed:** `train.py` (+28/-3), `configs/finetune_v11.yaml` (+95)

Critical insight: **validation loss is a bad checkpoint metric during DC warmup**. The DC loss grows monotonically as `dc_weight` ramps from 0 to  $\delta$  over N epochs. This means validation loss increases during the DC ramp phase regardless of how well the model is learning to classify, so `best_model.pth` was always frozen at epoch 20 (just before DC activates).

Switched checkpoint selection to **stenosis F1** as the primary metric:

- F1 is DC-immune — it measures classification quality directly
- Added F1-based patience counter for early stopping
- Added stenosis/plaque precision+recall + best\_stenosis\_f1 to TensorBoard

This single change was the primary reason v12 dramatically outperformed v11.

**467099a** | 2026-04-06 | *fix+docs: revert v11 DC hold to 20, add metrics doc and dataset script*

**Files changed:** `configs/finetune_v11.yaml` (+6/-2), `create_dataset.py` (+44), and 1 more (+307/-2)

`dc_warmup_hold=5` caused stenosis F1 to peak at 0.38 and then decline — the DC loss activated while the 6-class classification heads were still unstable post-initialization. Reverted to `hold=20` which was proven stable in v7-ft and v10-1D.

Also added `METRICS_2026-04-06.md` documenting all results across all versions and a `create_dataset.py` utility.

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**9b31187** | 2026-04-06 | *feat: add SC-Net pipeline flowchart visualization*

**Files changed:** `SC_Net_Pipeline_Flowchart.html` (+1210)

Created a comprehensive interactive HTML flowchart visualizing the entire SC-Net pipeline for documentation and presentation purposes.

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**c51a3e0** | 2026-04-06 | *feat: add v12 fine-tuning config (T0=60, patience=100) + v11 final results*

**Files changed:** `METRICS_2026-04-06.md` (+19), `configs/finetune_v12.yaml` (+97)

v11 post-mortem: T0=30 caused the LR to reach near-zero exactly at DC activation (epoch 20). The model never recovered — early stopping fired at epoch 61 with the best checkpoint from epoch 1 (a degenerate case).

v12 fixes both problems:

- `T0=60`: First cosine cycle ends at epoch 60, well past DC activation at epoch 20
- `patience=100`: Gives the model enough runway to surpass the initial pre-trained spike
- All v10-1D improvements retained: `boost_nonsig`, ordinal EMD, 1D IoU, F1 checkpoint metric

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## Phase 15 — v10/v11/v12 Training Run Improvements {#phase-15}

*(Covered by commits in Phase 14 above.)*

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## Phase 16 — v12 Best Results & Final Visualization {#phase-16}

**eefd63b** | 2026-04-07 | *docs+results: v12-ft final results — Stenosis F1=0.739, Plaque F1=0.502 (new best)*

**Files changed:** `calibration_thresholds_v12_constrained.json` (+48), `report.md` (+100), and 1 more (+188)

**New overall best results** for the project:

| Metric | v7-ft (prev best) | v12-ft (new best) | Change |
|--------|-------------------|-------------------|--------|
|--------|-------------------|-------------------|--------|

| Metric        | v7-ft (prev best) | v12-ft (new best) | Change |
|---------------|-------------------|-------------------|--------|
| Stenosis F1   | 0.585             | 0.739             | +26.3% |
| Stenosis ACC  | 0.580             | 0.736             | +26.9% |
| Sig Recall    | 0.595             | 0.733             | +23.2% |
| NonSig Recall | 0.581             | 0.639             | +10.0% |
| Plaque F1     | 0.463             | 0.502             | +8.4%  |

The key changes that drove this: native 1D IoU (correct geometry), F1-based checkpoint metric (correct model selection), T0=60 (stable LR during DC warmup), and patience=100 (full training runway).

Calibration thresholds: [H=2.80, NS=0.65, Sig=0.20].

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6da4793 | 2026-04-07 | feat: match visualize.py to Figure 3 exactly — dual bars (stenosis + plaque)

Files changed: visualize.py (+132/-45), .gitignore (+7)

Replaced the single combined 6-class bar with two thin bars per model row matching Figure 3 exactly:

- Top bar: stenosis severity (green=Healthy, yellow=Non-sig, orange=Significant)
- Bottom bar: plaque composition (green=None, blue=Calcified, pink=Non-calcified, purple=Mixed)

Added helper functions `_raw_to_sten_class()` and `_raw_to_plaque_class()` to extract the separate predictions from the combined 6-class model output. Legend colours corrected to match paper (yellow #FFC107, blue #2196F3).

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0fe587b | 2026-04-07 | feat: fix visualize.py bar layout to match Figure 3 exactly

Files changed: visualize.py (+99/-55)

Pixel-perfect layout fixes:

- Zero gap between stenosis/plaque bars within each model group
- Visible gap between model groups (GT, v12-ft, v7-ft)
- Legend in dedicated axes below the bars (not cramped into the GT bar)
- Manual absolute axes positioning via `fig.add_axes([x, y, w, h])` for precise control

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500d57d | 2026-04-07 | docs: add Phase 14b — dual-bar visualization fix matching Figure 3 exactly

Files changed: report.md (+61)

Documented the visualization overhaul: layout changes, colour corrections, and a full regeneration of 3182 visualization images with v12-ft vs v7-ft comparison.

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## Phase 17-19 — Architectural Improvements for v13 {#phase-17-19}

**fe9e990** | 2026-04-13 | *feat: add dc\_temperature annealing to dual\_task\_contrastive\_loss*

**Files changed:** `optimization.py` (+13/-4)

Added temperature annealing to the soft dual-task contrastive loss. In the DC soft mode, pseudo-labels are generated via `softmax(logits / T)`. With  $T > 1$ , the softmax is "softer" (more uniform probability distributions), and with  $T \rightarrow 1$ , it approaches the standard argmax.

Annealing  $T$  from 3.0 (soft/uncertain early in training) to 1.0 (sharp/confident late in training) follows the idea that pseudo-labels should be exploratory early on but become more decisive as the model matures.

**Research context:** Temperature annealing in knowledge distillation (Hinton et al., 2015) controls the "knowledge" transferred between teacher and student. High temperature emphasizes the dark knowledge (soft probabilities over non-predicted classes). This technique has been successfully applied in semi-supervised learning and self-distillation.

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**984b602** | 2026-04-13 | *feat: wire dc\_temperature annealing schedule in trainer*

**Files changed:** `train.py` (+13)

Wired the temperature annealing schedule into `train.py`: the temperature decreases linearly from `dc_temperature_start` (3.0) to 1.0 over the `dc_warmup_ramp` period. The temperature is passed to the loss function at each epoch.

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**864dba9** | 2026-04-13 | *feat: fix 2D/3D feature streams to be truly parallel (paper Fig.2)*

**Files changed:** `architecture.py` (+1/-1)

A subtle but impactful architectural bug. In the spatial branch's `feature_extraction_3d`, the extraction loop processes feature levels 0, 1, 2, 3. At level 0, the 2D extraction block correctly receives the input `x_2d = input`. However, at levels 1+, the 2D block was being fed `x_3d` (the output of the 3D extraction block) instead of its own previous `x_2d` output.

This meant both streams were processing the same 3D features after the first level — the 2D stream converged to the same representation as the 3D stream, completely defeating the purpose of having independent streams as shown in the paper's Figure 2. The fix is a single 1-word change: pass `x_2d` instead of `x_3d`.

**Impact:** Every training run up to and including v12 had this bug. The 2D and 3D streams were not independent. The SE fusion gate (Phase 18) learns to combine independent representations — it requires this fix to be meaningful.

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**b7ea87b** | 2026-04-13 | *feat: replace scalar \_3d\_weight fusion with per-level SE attention gate*

**Files changed:** `architecture.py` (+38/-2)

Replaced the fixed scalar blending weight (`_3d_weight=0.75` in all prior runs) with a learned per-level Squeeze-Excitation (SE) attention gate (`_FusionGate`).

#### How it works:

1. Concatenate 3D and 2D features:  $[B, 2C, D', H', W']$
2. Global average pool to channel descriptor:  $[B, 2C]$
3. Fully connected squeeze:  $\text{Linear}(2C \rightarrow C/2) \rightarrow \text{ReLU}$
4. Fully connected excitation:  $\text{Linear}(C/2 \rightarrow C) \rightarrow \text{Sigmoid} \rightarrow \alpha \in (0,1)$
5. Fuse:  $\alpha \cdot x_{3d} + (1-\alpha) \cdot x_{2d}$  per channel

Different channels can dynamically attend more to the 3D or 2D stream based on feature content. For example, calcified plaques (high HU, bright in CT) may be clearer in 2D cross-sections; volumetric context (lesion extent, shape) favors the 3D stream.

Adds ~220K new parameters. Old checkpoints still load (old `_3d_weight` parameter kept as a no-op for backward compatibility).

**Research context:** SE networks (Hu et al., 2018, "Squeeze-and-Excitation Networks", CVPR Best Paper) enable channel-wise feature recalibration at negligible computational cost. They have been widely adopted in medical imaging (e.g., SE-ResNet for chest X-ray classification). Applying SE gating to fuse complementary 2D/3D representations is a natural extension.

**c482707** | 2026-04-13 | *config: add v13 pre-train and fine-tune configs*

**Files changed:** `configs/pretrain_v13.yaml` (+93), `configs/finetune_v13.yaml` (+104)

v13 configuration incorporating all Phase 17–19 changes:

#### `pretrain_v13.yaml`:

- 300 epochs (vs 200 for v10) — the SE gate parameters need a full curriculum to learn meaningful channel routing
- Cosine warm restarts  $T_0=80$  — two full LR cycles in 300 epochs
- `soft_dc + dc_temperature_start=3.0` — soft pseudo-labels during pre-training

#### `finetune_v13.yaml`:

- Base: v12 best params ( $T_0=60$ ,  $\text{patience}=120$ ,  $\text{SWA}@100$ )
- $\text{lr}=2.5\text{e-}5$  (slightly lower than v12's  $3\text{e-}5$  to protect the newly-trained gate weights)
- `dc_temperature` annealing  $3.0 \rightarrow 1.0$  over ramp epochs 20–60
- All v12 improvements retained: `boost_nonsig`, ordinal EMD, 1D IoU, F1 checkpoint, soft DC, focal loss, balanced sampling

**75855de** | 2026-04-13 | *docs: add Phase 17–19 and v13 config summary*

**Files changed:** `report.md` (+200)

Comprehensive documentation of all three Phase 17–19 improvements:

- Phase 17: True parallel 2D/3D streams — root cause analysis, the 1-word fix, expected impact
- Phase 18: SE attention-based view fusion — `_FusionGate` class design, ~220K new parameters, channel routing intuition
- Phase 19: DC temperature annealing — implementation in `optimization.py` and `train.py`, schedule design

Metrics Progression Summary {#metrics-summary}

| Version             | Stenosis F1 | Stenosis ACC | Sig Recall | NonSig Recall | Plaque F1 | Key Change                                      |
|---------------------|-------------|--------------|------------|---------------|-----------|-------------------------------------------------|
| v1 (pretrain)       | 0.413       | 0.702        | —          | —             | 0.100     | Baseline                                        |
| v6-ft (calibrated)  | 0.393       | 0.435        | 0.553      | —             | 0.181     | First fine-tune, calibration                    |
| v7-ft (constrained) | 0.585       | 0.580        | 0.595      | 0.581         | 0.463     | DC warmup, confidence gating, balanced sampling |
| v8-ft               | 0.555       | —            | —          | —             | —         | focal_gamma=3.0 (worse — ablation finding)      |
| v9-ft               | 0.643       | 0.645        | 0.456      | 0.456         | 0.488     | Ordinal EMD, SWA, warm restarts                 |
| v12-ft              | 0.739       | 0.736        | 0.733      | 0.639         | 0.502     | Native 1D IoU + F1 checkpoint metric            |

Total improvement: Stenosis F1 +79% relative, from 0.413 to 0.739

Summary of Research Areas Covered

1. **Architecture** — DETR-style object detection, 3D CNN + Transformer, 2D/3D hybrid feature extraction, SE attention gates, learnable positional encoding
2. **Loss functions** — Focal loss, ordinal EMD loss, dual-task contrastive loss, soft pseudo-labels, temperature annealing, class-weighted CE
3. **Training techniques** — AMP, DDP, EMA, layer-wise learning rates, cosine LR scheduling with warm restarts, SWA, gradient accumulation, balanced sampling, early stopping
4. **Evaluation** — Threshold calibration (standard and constrained), TTA, ensemble inference, per-class metrics, AUC-ROC, confusion matrices
5. **Explainability** — 3D Grad-CAM, Monte Carlo Dropout uncertainty estimation
6. **Data** — Multi-window CT normalization, patient-level splitting, clinically credible augmentation, new dataset format support
7. **Infrastructure** — TensorBoard, YAML configs, comprehensive evaluation pipeline, CPR visualization matching paper Figure 3